

ICAIMT - AI Assistant for Wealth Management «ECO InvestMind AI»

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Abstract: Managing personal finances remains a significant challenge in developing economies, where access to financial advisory services is limited and financial literacy is often low. In response to this, we propose «*ECO InvestMind AI*», an intelligent mobile application designed to democratize wealth management through the integration of advanced machine learning (ML) and natural language processing (NLP) techniques. Our system leverages a dual-stream hybrid architecture combining a fine-tuned BERT model for sentiment analysis of financial news and an MLP-based time series model for stock price forecasting. The application provides personalized stock recommendations by analyzing market trends, ESG scores, and sentiment signals, offering users real-time, behaviorally adaptive financial guidance. To ensure scalability and maintainability, the app is built using Clean Architecture principles and Kotlin Multiplatform, enabling seamless cross-platform performance. A relational database structure supports the integration of diverse datasets, including historical stock prices, ESG metrics, and sentiment-labeled news. Evaluation results demonstrate high predictive accuracy across multiple Kazakhstani companies and indices. This paper illustrates how AI-powered tools like «*ECO InvestMind AI*» can bridge the gap between traditional investment tools and modern financial advisory systems, empowering users with actionable insights and promoting inclusive participation in financial markets.

Keywords: Wealth management, sentiment analysis, stock forecasting, BERT, MLP, Kotlin Multiplatform, ESG, mobile finance, Kazakhstan

1. Introduction

Managing personal finances remains a complex challenge for many individuals, particularly in developing economies. Professional financial advisors can be expensive, rendering their services inaccessible to much of the population. This issue is especially pronounced in Kazakhstan, where, despite notable economic growth, financial literacy levels remain relatively low. A recent study reported that nearly 30% of Kazakhstani adults with limited financial knowledge have never invested, limiting their ability to accumulate long-term financial security and resilience (Bayakhmetova, Kadyrzhanov, & Aitzhanova, 2023).

Many people, especially young people, focus on building their careers, pursuing education, or taking care of their health, yet they often overlook the importance of managing money wisely. Lack of financial knowledge can lead to poor budgeting, mounting debt, or even falling victim to fraudulent schemes. For example, if someone does not have a clear understanding of their income and expenses, they may end up in a difficult situation that affects not only their personal finances but also the broader economy.

At the same time, many Kazakhstani people continue to rely on traditional bank deposits as their primary savings strategy. By December 2024, total individual deposits in Kazakhstani banks had reached 24.5 trillion tenge, a 23.4% increase compared to the previous year (Zakon.kz, 2024). While such deposits offer a sense of financial stability, they often fail to outpace inflation.

As noted by Timur Turlov, a prominent Kazakhstani entrepreneur, inflation continues to exert a strong influence on the country's economy, leading to the steady depreciation of money. He argues that, in a period of persistent inflation, relying on low-yield instruments such as government bonds or bank deposits is a weak strategy. "Those who seek maximum peace of mind may become the most vulnerable," he warns, highlighting that such approaches could result in a gradual loss of purchasing power (Turlov, 2025).

Faced with these limitations, individuals are turning to digital platforms to diversify and grow their savings. In recent years, investment applications such as Tabys, Freedom24, Jusan Invest, and Halyk Invest have broadened market access, offering lower entry thresholds and mobile-first experiences. However, these platforms often provide only basic functionality—such as asset listings or price charts—and lack personalization or adaptive guidance. Moreover, few utilize machine learning (ML) or sentiment analysis, despite growing evidence that market sentiment significantly influences investor behavior.

In this article, we will explore how modern mobile technologies and machine learning algorithms are transforming traditional approaches to asset management. We begin by examining the current challenges in financial services and then show how innovative solutions can make financial planning more effective and easier to understand for a wide range of users, exploring how ML techniques can enhance wealth management applications by addressing the personalization gap in existing platforms.

Also, we introduce «ECO InvestMind AI», a mobile application designed to deliver real-time, with a new approach of personalized investment recommendation. By incorporating sentiment analysis of financial news and predictive modeling, the platform adapts to users' financial goals, behavioral patterns, and emotional contexts. Ultimately, this paper demonstrates how AI-driven tools can bridge the financial literacy gap, promote more inclusive participation in financial markets, and empower users to manage their economic future more effectively.

2. Literature review

2.1 The Role of Financial Literacy and Behavior in Investment Decision-Making

Effective personal finance management depends on more than just access to investment tools; it requires an ability to understand, evaluate, and act upon financial information. In Kazakhstan, low financial literacy has emerged as a critical barrier to participation in investment markets. Bayakhmetova et al. highlights that individuals with poor financial understanding are significantly less likely to invest, regardless of income or age (Bayakhmetova, Kadyrzhanov, & Aitzhanova, 2023).

Compounding this issue is the proliferation of information in digital media. Inexperienced users, overwhelmed by conflicting news and social media signals, often make emotionally driven decisions—such as panic-selling during market volatility or abandoning long-term investment plans prematurely. Financial education alone may not be sufficient to counteract these behaviors unless accompanied by real-time, context-aware decision support.

Research in personal finance emphasizes the behavioral dimension of money management. Zhigitbek argues that building wealth begins with disciplined habits, such as budgeting, tracking expenses, and consistently saving a portion of one's income (Zhigitbek, 2022). Samuel in his widely cited work *The Richest Man in Babylon*,

similarly, underscores the importance of setting aside at least 10% of earnings, regardless of income level. These principles point to a critical insight: financial technologies should not only offer market access but also reinforce sound decision-making behavior (Samuel, 2021).

A user-friendly interface, straightforward asset selection, and clear investment mechanisms enable beginners to take their first steps in capital management without unnecessary complications. This is especially important given the low level of financial literacy observed in many regions of Kazakhstan. In such areas, people often fall victim to financial pyramids while trying to invest and grow their capital. To help address this issue, the Astana International Exchange (AIX), with support from the National Bank of Kazakhstan, launched the Tabys app—a social initiative aimed at improving financial literacy and expanding access to investment tools. This app aims to improve investment accessibility and financial literacy in Kazakhstan. (GlobeNewswire, 2025) It allows users to start investing from as little as \$1, offering access to various securities, including exchange-traded notes and gold coins issued by the National Bank of Kazakhstan. Tabys also features educational materials to guide novice investors (Tabys App – Official Website, n.d.).

2.2 Existing Digital Platforms in Kazakhstan

At this stage, investors can utilize mobile applications provided by banks and investment platforms such as Freedom Broker, Jusan Invest, and Halyk Invest, which allow convenient trading of local stocks and bonds. These applications enable users to open brokerage accounts directly from their smartphones, eliminating the need for physical office visits.

For instance, financial management tools like CoinKeeper enable users to track their daily budgets and expenses in real time. This application provides visual tools for creating a financial plan, saving money, and managing expenses efficiently (CoinKeeper, 2023).

Freedom24, launched in 2020 by Freedom Finance, provides users with access to over 1 million financial instruments, including stocks, ETFs, bonds, and options from global exchanges. The platform offers professional market analysis and investment ideas, aiming to assist users in making informed decisions (Freedom24, n.d.).

Jusan Invest, part of the Jusan Super App, offers a comprehensive suite of financial services, including the ability to open brokerage accounts, purchase and sell shares, and access analytics and investment ideas. The app integrates banking, shopping, insurance, and mobile communications, providing holistic financial management experience (Jusan Invest, n.d.).

Another mobile application, Halyk Invest, enables online trading of equities and bonds of leading Kazakhstani and international companies. The app provides users with access to ETFs, market news, and analytics from Halyk Finance's research team, facilitating informed investment decisions (Halyk Invest, n.d.).

While these platforms have democratized access to financial markets, especially for young or first-time investors, they often lack mechanisms for real-time personalization or intelligent financial recommendations. Investment suggestions, if present, tend to be static or based on simplistic portfolio allocation rules. Moreover, sentiment analysis—a method of interpreting the emotional tone of financial news and social media—is absent in almost all Kazakhstani apps, despite its potential to contextualize market risks and investor behavior.

Despite their progress, most local platforms lack mechanisms for real-time personalization or intelligent financial recommendations. Investment suggestions, if present, tend to be static or based on simplistic portfolio allocation rules. Moreover, sentiment analysis—a method of interpreting the emotional tone of financial news and social media—is absent in almost all Kazakhstani apps, despite its potential to contextualize market risks and investor behavior.

While CoinKeeper focuses on budgeting and Tabys emphasizes accessibility, none of these platforms fully

integrate ML or LLM-based reasoning to support decision-making. This leaves a clear opportunity for more advanced, AI-enhanced applications that can help users interpret complex market signals and align investment strategies with personal goals.

2.3 AI-driven financial platforms

Recent advancements in artificial intelligence (AI) and machine learning (ML) have significantly transformed the financial technology landscape, particularly in the domain of automated wealth management and financial advisory services. Several state-of-the-art AI-driven financial platforms have emerged, incorporating large language models (LLMs), retrieval-augmented generation (RAG), and deep learning architectures to provide sophisticated market analysis, predictive modeling, and portfolio management recommendations. Among the most notable of these are AlphaFin, FinRobot, and FinGPT, each of which leverages advanced computational techniques to enhance financial decision-making. However, despite their innovations, these existing frameworks exhibit certain limitations in terms of personalization, explainability, and real-time adaptability, which our proposed financial advisor seeks to address.

AlphaFin represents a pioneering effort in integrating RAG techniques with financial forecasting, aiming to improve stock market trend predictions and financial question-answering capabilities. The AlphaFin dataset is a comprehensive collection that combines traditional financial datasets, real-time market data, and manually curated Chain-of-Thought (CoT) reasoning data. This hybrid approach enhances both the interpretability and robustness of machine learning models, allowing for more transparent and justifiable financial analyses. (Li et al., 2024)

One of the key advantages of AlphaFin is its explicit reasoning pathway, which enables LLMs to generate not only predictions but also logical justifications for investment decisions. Furthermore, the inclusion of real-time financial data ensures that models trained on AlphaFin remain relevant and responsive to dynamic market conditions. However, while AlphaFin significantly advances the domain of AI-driven financial analytics, it faces notable challenges, including the high computational cost of real-time data integration, potential biases inherent in financial datasets, and the limited scope of its advisory capabilities, which remain largely focused on stock trend predictions rather than holistic wealth management strategies. (Li et al., 2024)

FinRobot is an open-source AI agent platform designed to address various financial analytical challenges by employing a modular, multi-agent architecture that integrates specialized LLMs for different financial tasks. Unlike AlphaFin, which primarily focuses on stock trend prediction, FinRobot extends its functionalities to equity research, financial document analysis, and risk assessment, making it a versatile tool for both institutional and retail investors (Yang et al., 2024)

At its core, FinRobot consists of four key layers: (1) the Financial AI Agents Layer, which employs Financial Chain-of-Thought (CoT) prompting to decompose complex financial problems into logical, interpretable steps; (2) the Financial LLM Algorithms Layer, which dynamically configures model applications for specific financial tasks; (3) the LLMops and DataOps Layer, which optimizes the processing of financial data and enhances model accuracy through continuous fine-tuning; and (4) the Multi-Source LLM Foundation Models Layer, which ensures flexibility by integrating multiple state-of-the-art LLM architectures for financial analysis (Zhou, Wang, Wu, & Yang, 2024)

While FinRobot significantly enhances financial research through its structured analytical framework, it exhibits several limitations. First, despite its adaptability, its reliance on a multi-agent CoT system may lead to computational inefficiencies, particularly when processing large-scale financial datasets. Second, its applicability to global financial markets remains constrained by regional data availability and the inherent difficulty of adapting AI-driven financial models to the nuances of different regulatory environments. Third, FinRobot, like many LLM-based financial systems, struggles with real-time market fluctuations and often requires external validation to ensure the accuracy of its insights (Zhou, Wang, Wu, & Yang, 2024)

FinGPT represents a specialized adaptation of general purpose LLMs for financial applications, leveraging

domain-specific fine-tuning to enhance the model’s ability to process financial news, market reports, and investor sentiment data (Yang, Liu, & Wang, 2023). This model is designed to improve financial forecasting and advisory services by integrating a wide array of financial datasets into its training process, thereby enabling it to generate context-aware recommendations.

One of the primary advantages of FinGPT is its capacity for text-based financial analysis, which allows it to process financial news, earnings reports, and investor commentary with a high degree of accuracy. However, its reliance on textual data poses inherent limitations, particularly in scenarios requiring real-time quantitative analysis, market risk assessment, or multi-modal data integration. Additionally, while FinGPT excels in sentiment-driven financial forecasting, it lacks a robust framework for personalized wealth management and investment diversification strategies (Yang, Liu, & Wang, 2023)

There are plenty techniques to optimize the function to get the best solution in decision-making systems. The most popular of them is Particle Swarm Optimization (PSO). Many papers were written about researching this techniqua (Figure 1).

This approach with Large Language Models (LLMs) presents a promising frontier in solving complex, multi-dimensional problems. While traditional PSO techniques have proven effective in diverse areas—from feature selection to resource scheduling and decision-making under uncertainty—they often face limitations such as premature convergence and challenges in scalability (Kennedy & Eberhart, 1995; Mahmoudi & Feylizadeh, 2018).

LLMs have the potential to enhance PSO's interpretability and contextual adaptability. For example, LLMs could aid in the interpretation of Nash equilibrium adjustments (Bao et al., 2020), improve decision traceability using Shapley values (Lounis et al., 2022), and provide semantic understanding in hybrid PSO applications (Altybayeva, 2023). However, practical limitations remain, particularly in high-dimensional search spaces, real-time optimization, and model explainability (Dökeoğlu et al., 2022; Altybayeva, 2023).

Study	Technical Description	Role of Large Language Models	Challenges
Dökeoğlu T., Deniz F. A., Kızıltöz H. (2022) "A Comprehensive Survey on Recent Metaheuristics for Feature Selection" Neurocomputing, Vol.494, 269-296.	A comprehensive survey reviewing over 20 years of feature selection algorithms. The study categorizes feature selection methods into filter, wrapper, and embedded approaches and highlights the use of metaheuristic algorithms (e.g., PSO, Genetic Algorithms, Firefly Algorithm). The wrapper approach is identified as the most effective but computationally expensive.	While not directly related to LLMs, these algorithms could enhance feature selection for LLM-powered systems by optimizing high-dimensional inputs and reducing computational overhead.	Addressing computational scalability when feature selection is applied to datasets with over 1 million features or instances. Risk of overfitting when using wrapper approaches in small datasets.
Kennedy J., Eberhart R. (1995) "Particle swarm optimization" Proceedings of ICNN'95 - International Conference on Neural Networks, Perth, WA, Australia, 1995, pp. 1942-1948 vol.4.	The foundational work on Particle Swarm Optimization (PSO) introduced a method inspired by the social behavior of bird flocks and fish schools. PSO operates by iteratively improving candidate solutions based on individual and collective experiences of particles. Tested successfully in multi-dimensional optimization problems.	LLMs could act as a guide in initializing the PSO particles by providing contextual insights about the problem space. Additionally, LLMs can assist in the interpretation of results for complex real-world optimization scenarios.	PSO struggles with premature convergence in search spaces with more than 10 dimensions, often requiring hybrid solutions or adjustments such as dynamic inertia weighting.
Bao T., Huynh Q., Nguyen X., Nhu N., Le D. (2020) "A Novel Particle Swarm Optimization Approach to Support Decision Making in the Multi Round of an Auction by Game Theory" International Journal of Computational Intelligence Systems, Vol. 13, 1447-1463.	Proposed a novel hybrid approach combining PSO with Nash equilibrium principles from game theory. Demonstrated its effectiveness in optimizing decisions for multi-round auction problems. The study evaluated performance metrics such as optimality and convergence on datasets with up to 1000 participants and 10 bidding rounds.	LLMs can enhance this hybrid approach by identifying patterns in participant behavior or by forecasting future rounds. This can lead to more accurate contextual adjustments in the Nash equilibrium calculations.	High computational cost for complex scenarios involving many participants and rounds. Challenges arise in modeling realistic agent behavior that aligns with theoretical equilibrium states.
Mahmoudi A., Feylizadeh M. (2018) "A novel algorithm for solving resource constrained project scheduling problems: a case study" Journal of Advances in Management, Vol. 15, Louiche M., Nesmaoui R., Mbarek M., Lazaar M. (2023) "Shapley Values for Explaining the Black Box Nature of Machine Learning Model Clustering" Procedia Computer Science, Vol. 220, 806-811	Applied PSO to resource-constrained project scheduling problems, addressing dynamic resource allocation. The model showed efficiency in scheduling tasks within tight deadlines and budgets. The study used datasets with up to 50 tasks and 10 resource constraints. Leveraged Shapley values to explain the decision-making process of black-box machine learning models. Demonstrated on clustering algorithms with datasets containing up to 100,000 instances, emphasizing the importance of explainability in critical applications such as financial risk assessment and healthcare.	LLMs could assist in contextualizing dynamic constraints (e.g., shifting priorities in real-time) and provide better interpretation of output schedules for decision-makers. LLMs can benefit from Shapley values to interpret decisions, especially in contexts where transparency is crucial (e.g., explaining why a certain feature was deemed important in an optimization result).	PSO's susceptibility to local optima is especially problematic in resource allocation problems with dynamic constraints. Requires hybridization (e.g., with Genetic Algorithms) for more robust solutions. Trade-off between computational cost and level of explainability, particularly when dealing with large-scale datasets or complex machine learning models (e.g., neural networks with thousands of parameters).
Altybayeva N. (2023) "Optimization of decision making process in project management framework using hybrid method of particle swarm optimization and game theory" The Twenty-Ninth International Symposium on Artificial Life and Robotics 2024 (AROB 29th 2024)	Proposes a hybrid approach combining Particle Swarm Optimization (PSO) with game theory principles to optimize decision-making in project management. The model considers risk assessment, resource allocation, and team collaboration to achieve optimal outcomes. It was tested on multidimensional problems.	LLMs could enhance the contextual understanding of project management challenges, guide the initialization of PSO particles, and provide explanations for decisions derived using game-theoretic models.	Combining PSO and game theory adds computational complexity, especially for large-scale, multi-dimensional decision-making problems. Integration into dynamic real-world scenarios remains challenging.

Figure 1. Comparative Analysis of Studies Integrating Particle Swarm Optimization (PSO) and Large Language Models (LLMs): Technical Descriptions, Applications, and Challenges

Studies suggest that hybrid models combining PSO and LLMs may significantly enhance optimization outcomes, particularly in scenarios requiring adaptive reasoning and complex decision logic (Lounis et al., 2022; Altybayeva, 2023). Yet, the challenge of scaling such systems and integrating real-world data with theoretical models continues to limit widespread application.

3. «ECO InvestMind AI»: A recommendation system for investing in stocks

3.1 Limitations of Existing Solutions and Our Proposed Approach

While existing AI-driven financial advisory platforms such as AlphaFin, FinRobot, and FinGPT have made significant contributions to the field of automated wealth management, they remain limited in their ability to provide fully personalized, comfortable, real-time, and behaviorally optimized financial planning solutions. Our proposed financial advisor represents a novel approach that bridges these gaps by integrating machine learning models with highly customized user experience, thereby offering a comprehensive and adaptive financial advisory platform tailored to individual users' evolving needs.

Investing in the financial markets involves significant risks, largely due to volatility and the influence of diverse external factors, including public sentiment driven by financial news. Traditional investment decision-making often overlooks or inadequately integrates the emotional and informational impact of news events, potentially leading to suboptimal financial decisions. Therefore, there is a need for advanced tools capable of accurately predicting stock prices by effectively incorporating both quantitative market data and qualitative sentiment analysis of news sources.

Although the AI-driven financial advisory systems exhibit significant advances in the application of machine learning to financial analysis, they fall short in several critical areas, including personalized financial planning, risk-adjusted portfolio recommendations, and real-time adaptive forecasting. Our proposed financial stock recommender, «ECO InvestMind AI», aims to address these gaps by integrating cutting-edge ML algorithms with highly personalized user experience.

Key innovations of our proposed system include:

(1) advanced ML-driven investment diversification and nudging mechanisms, which is integrated in our platform integrates ML-based diversification strategies that optimize asset allocation in accordance with market conditions and user preferences; (3) real-time income and expense forecasting with visualization, which will help the financial advisor to generate predictive insights on future income and expenses, allowing users to make informed financial decisions; (4) cost optimization and recommender system for expense reduction, that identifies opportunities for cost savings, such as optimizing subscription services, selecting more cost-effective financial products, or adjusting lifestyle expenditures to align with savings goals.

Our objective is to develop a mobile stock recommendation system that leverages machine learning (ML) algorithms to provide precise stock price predictions. This application will uniquely integrate sentiment analysis derived from financial news, thus offering enhanced, context-aware investment recommendations. By combining quantitative stock market trends and qualitative news sentiment, the proposed application aims to deliver personalized, actionable insights to investors, improving decision-making quality and ultimately financial outcomes.

3.2 Software Architecture

The financial recommender application is structured according to the principles of Clean Architecture, which promotes modularity, maintainability, and scalability. It is organized into three primary layers that work together to ensure a robust and flexible software system. Its software application can be seen in Figure 2.

At the forefront is the Presentation Layer, which is developed using Android's Jetpack Compose. This layer is

responsible for managing user interactions through an intuitive interface and is connected to the ViewModel. The ViewModel facilitates the presentation of real-time updates using LiveData, ensuring seamless and responsive user experiences.

The Domain Layer contains the core business logic of the application. It is centered around well-defined use cases that encapsulate key functionalities, such as generating stock recommendations via the StockRecoUseCase and listing available stocks through the StockListUseCase. This separation of concerns enhances code clarity and testability, supporting the application's long-term maintainability.

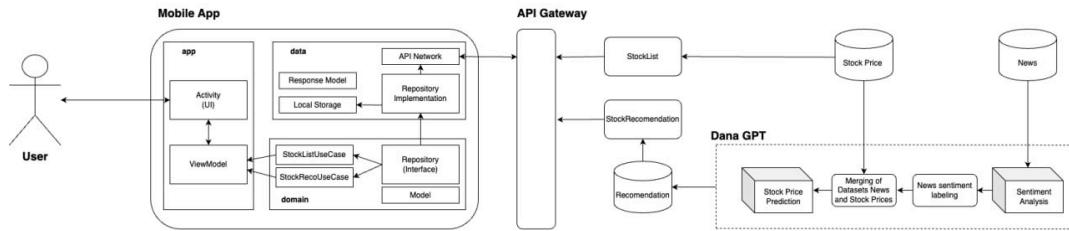


Figure 2. ECO Invest AI Software Architecture

Finally, the Data Layer manages data acquisition, processing, and persistence. It includes an API network component designed to fetch real-time financial information from external sources. Additionally, local storage mechanisms enable data caching and offline access, ensuring that user data remains synchronized even when the application is used without an internet connection. The repository implementation serves as an intermediary between the data sources and the domain use cases, establishing a clean and consistent data flow across the application.

Our solution employs a layered enterprise architecture composed of three primary layers: the Presentation Layer, Business Logic Layer, and Data Layer (Figure 3).

The Presentation Layer, situated at the top of the architecture, interacts directly with the users through a user-friendly interface designed using Android Jet- pack Compose. This layer comprises two main components. The User Activity component provides an intuitive graphical user interface, enabling users to access key application features effortlessly. Additionally, the View Model acts as an intermediary, processing and transforming data received from the Business Logic Layer into observable states, thus facilitating reactive and real-time user interaction.

The Business Logic Layer houses the core functionality and rules of the Robo-advisor. It includes the Stock-RecoUseCase, responsible for generating personalized stock recommendations based on predictive models, and the StockListUseCase, which manages operations related to fetching and presenting lists of available stocks to users. This layer also features a Repository Interface, which abstracts data access, allowing seamless retrieval without dependency on specific data sources, and the Model component, which applies machine learning algorithms to provide accurate stock price predictions and sentiment-based insights.

The Data Layer manages and prepares data, ensuring its readiness for processing and analysis. It consists of Database Tables storing historical stock price data, ESG scores, and news sentiment analysis results. Additionally, Store Process/Views facilitate essential database operations and maintain updated views for efficient querying. The Data Preprocessing component is responsible for cleaning, transforming, and preparing raw data sourced from CSV files for machine learning tasks. Finally, Data CSV Files contain initial datasets such as historical stock prices, ESG scores, and news sentiment data utilized in training and analysis.

This layered architectural approach ensures clear separation of concerns, enhances system modularity,

facilitates easier maintenance and testing, and provides the scalability necessary to meet evolving user demands and increasing data complexities.

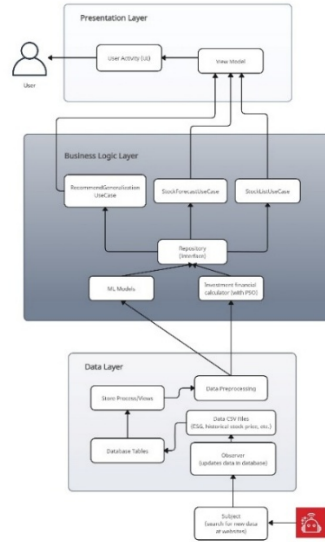


Figure 3. ECO Invest AI Layered Enterprise Architecture

A recommendation system for investing in stocks collects historical stock price data, news, and ESG information from large companies represented on the Kazakh and American stock markets. The data comes from web portals kapital.kz, kase.kz, finance.yahoo.com, as well as from the official websites of the companies themselves (to obtain basic information such as financial statements, company structure and strategic goals).

Additionally, ESG scores and ratings are downloaded from specialized platforms such as lseg.com. Analyzing the data obtained using artificial intelligence and machine learning methods, the program predicts stock price changes for the future 7 days and generates personalized recommendations for the user to buy or sell shares.

The Bot at the bottom of the diagram updates the data in the datasets every week as new data appears on the news websites and new stock prices are collected on the appropriate site.

3.3 Mobile Application Architecture

An application architecture of our mobile application based on Kotlin Multiplatform (KMP) is shown at Figure 4. Our goal is to create a single, reusable codebase that works efficiently across both Android and iOS platforms, reducing code duplication while maintaining native user experiences.

The core of the application is built upon a shared Kotlin module, which serves as the foundation for both Android and iOS platforms. This shared module includes clearly defined use cases that encapsulate the core application logic and manage user interactions. A unified repository layer is implemented to handle data from both remote and local sources, utilizing the Ktor library for seamless network communication and SQLite for reliable offline data storage. This architecture promotes consistency and reusability across platforms.

To ensure predictable and consistent user interface behavior, the application adopts the Model-View-Intent (MVI) architectural pattern for UI state management. This approach supports unidirectional data flow, which enhances the maintainability and testability of the UI logic.

For platform-specific features such as GPS, camera access, and sensor data, the application leverages Kotlin's

expect/actual mechanism. This interoperation layer allows clean and efficient integration of the shared Kotlin logic with native Android and iOS APIs, maintaining platform fidelity while maximizing code reuse.

On the native side, the Android interface is developed using Jetpack Compose, which enables the construction of modern and responsive user interfaces. For iOS, SwiftUI is utilized, ensuring full integration with native iOS SDK components. When a user interacts with the application, the platform-specific UI communicates with the shared module via the interop layer. The resulting data and logic flow back to the native UI, ensuring a smooth and responsive user experience. This design approach enhances development efficiency, simplifies maintenance, and leverages the unique capabilities of each platform.

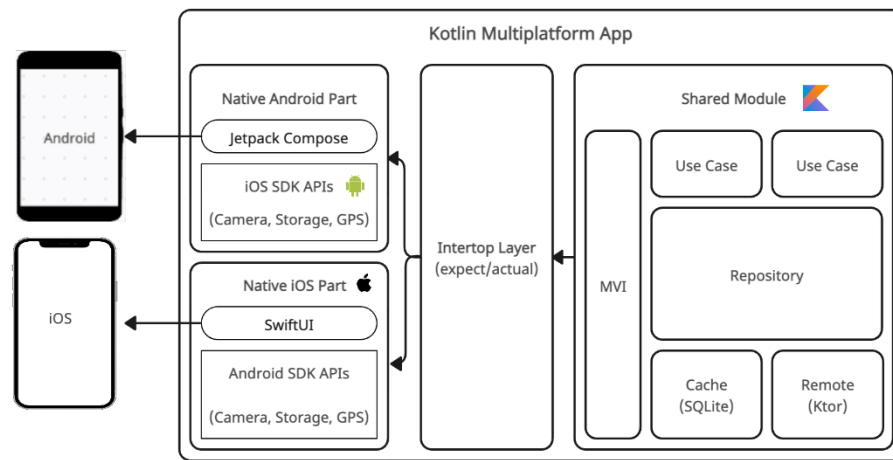


Figure 4. Kotlin Multiplatform Application Architecture

The primary scope of the application lies in providing a stock investment recommendation system designed for use in financial markets. It delivers detailed company information and offers buy or sell recommendations based on stock price forecasts generated through machine learning algorithms. The system automatically collects the latest financial news by parsing content from web portals such as Kapital.kz and Finance.Yahoo.com. It employs a BERT-based neural network, utilizing transformer architecture for sentiment analysis to determine the emotional tone of each article—categorizing content as positive, neutral, or negative.

Based on the sentiment analysis results, the application utilizes a regression-based neural network to forecast short-term stock prices. These forecasts are then used to generate informed investment recommendations, enabling users to make better financial decisions. The methodology integrates a structured data pipeline, advanced ML algorithms, and a mobile application framework to deliver a personalized financial advisor platform. This approach incorporates database design, data processing workflows, and software architecture principles to ensure robustness, scalability, and user-centric functionality.

3.4 Database Design and Dataset Description

The database, shown at Figure 5, designed to support the financial recommender application integrates heterogeneous data sources to enable comprehensive investment decision-making. The schema is organized around three primary data domains—stock market prices, financial news sentiment, and Environmental, Social,

and Governance (ESG) metrics—complemented by static company attributes and predictive model outputs. This relational database structure facilitates complex analytical workflows by linking time-series financial data with textual and sustainability indicators.

Stock price data is stored in a series of normalized tables representing different markets and individual tickers, such as KASE, NASDAQ, S&P 500, and specific company stocks including KSPI, HSBK and KZAP. Each stock price table contains daily records with attributes such as open, close, high, low, and volume. The combination of Ticker and Date serves as the composite primary key, allowing precise temporal alignment and historical tracking across exchanges. These price datasets were sourced from finance.yahoo.com, covering the period from 2020 to 2025.

Each stock price table is linked to a corresponding news table through foreign key relationships on the Ticker and Date fields. These news tables contain textual content related to the associated company or market, the date of publication, and sentiment labels assigned through automated sentiment analysis algorithms. The sentiment label is stored as an integer value, reflecting whether the tone of the news is negative, neutral, or positive. This alignment between news sentiment and trading data enables joint analysis of market movements in response to narrative signals, enhancing predictive capabilities.

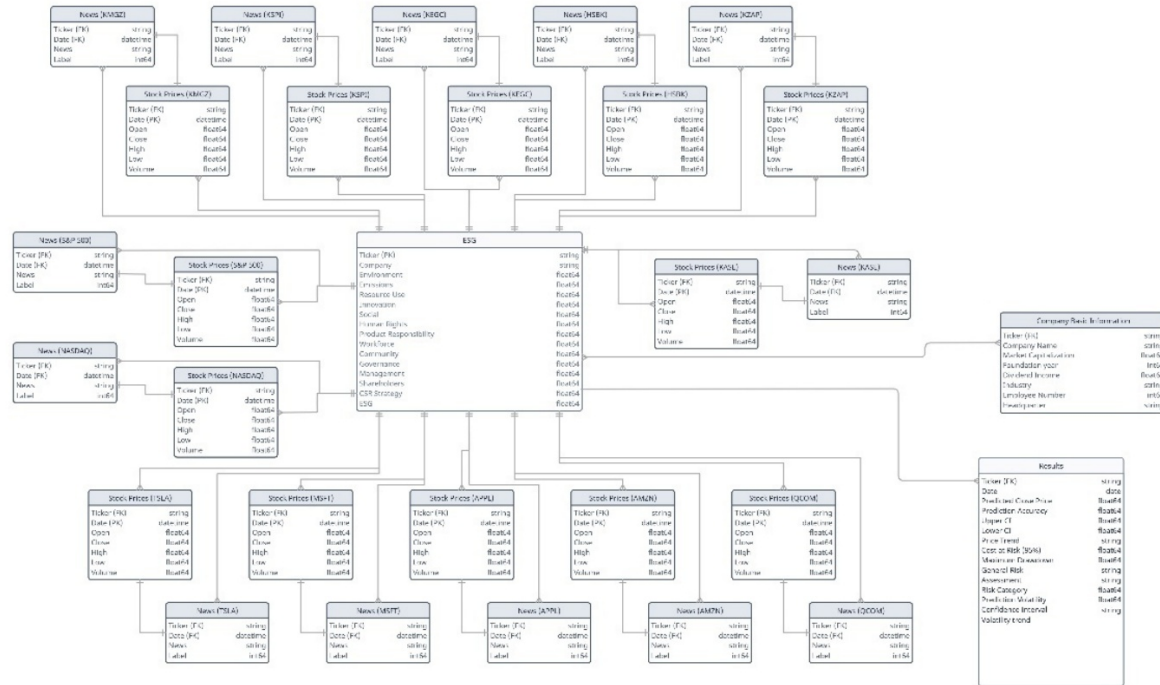


Figure 5. ECO Invest AI Database ER Diagram

In addition to market and sentiment data, the schema includes a dedicated ESG table that aggregates sustainability-related metrics for each company, indexed by the Ticker field. The table captures a wide range of environmental, social, and governance indicators including emissions, innovation, product responsibility, human rights, shareholder engagement, and overall ESG scores. These quantitative metrics provide a foundation for evaluating firms based on ethical and long-term risk criteria. The ESG table is structurally connected to the stock price and company information tables, creating a unified profile for each listed entity that spans financial and non-financial dimensions.

Static firm-level attributes are stored in the Company Basic Information table, which includes fields such as company name, market capitalization, industry classification, foundation year, dividend income, employee count, and headquarters location. This table shares the Ticker key with other core tables, providing a metadata layer for contextualizing financial performance and sustainability assessments.

Forecasting results generated by the system's machine learning models are captured in the Results table. Each record contains a predicted close price, prediction accuracy score, confidence intervals (upper and lower bounds), forecasted price trend, cost at risk, maximum drawdown, volatility indicators, and general risk assessments. These prediction outputs are tied to specific tickers and dates, aligning them with historical prices and sentiment events for validation and application. The data feeding these models is drawn from the integrated ecosystem of price, news, and ESG information, enabling multivariate time-series prediction and scenario analysis.

The relational model ensures referential integrity through consistent use of foreign key constraints, establishing direct links between time-indexed financial data and corporate-level features. News entries are aligned with corresponding stock price entries via shared date and ticker identifiers. ESG and company information are mapped to price and prediction records through ticker-based joins, allowing longitudinal and cross-sectional analyses of firm behavior. This architecture supports advanced analytics and real-time recommendation services by facilitating unified access to historical patterns, market sentiment, sustainability evaluations, and forecast data within a single, scalable data infrastructure.

3.5 Backend structure

User module basically contains user features. User feature is related to work with user model. Auth module has own validations, roles guard logic and controllers, but uses user features as basic. Auth works as security level for access to user. Financial module contains stocks and news. It scraps data in list of websites. Architecture, dividing all features by module is named module architecture.

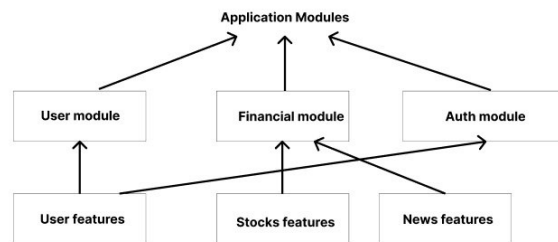


Figure 6. Backend Structure

Node.js frameworks basically use module architecture as default. Module specifies DTOs, entities, services, controllers and features as a directory. Usually separation by module belongs to data model. Module architecture allows reuse the code and convenient for all stages of development. DTO is expecting data format in HTTP request, usually it is used for validation of requests. Entity is instance of model or table of database.

Service is business logic of module. It can contain logical functions, work with database, validations, formulas. Controller is responsible for routing, requests, responses and other works with HTTP requests. Features can be used in other modules which allows reuse the code

4. AI Assistant based on Advanced Machine Learning models

4.1 Hybrid ML Architecture Overview

The core of the financial advisor system is a hybrid machine learning architecture named Dana, specifically designed to provide accurate and context-aware stock price forecasts for companies. The architecture integrates time series forecasting and natural language processing (NLP) techniques to incorporate both quantitative market data and qualitative financial sentiment into its predictive modeling.

The model architecture is organized into a dual-stream pipeline. The first stream focuses on sentiment analysis. Financial news headlines are collected, preprocessed, and passed through a fine-tuned BERT-based transformer model that performs sentiment classification. Each news item is assigned a sentiment label—positive, neutral, or negative—reflecting the emotional tone of the content. These sentiment labels are then aligned temporally with corresponding financial series data to ensure consistency in downstream analysis.

The second stream involves time series forecasting based on historical market data. Specifically, the system utilizes OHLCV (Open, High, Low, Close, Volume) features from KASE-listed stocks and feeds them into a MLP neural network. MLP is selected due to its proven ability to capture sequential dependencies and model market trends, volatility, and fluctuations over time.

The outputs of the sentiment analysis and MLP forecasting streams are subsequently integrated to produce the final predictive output. This fusion approach allows the model to dynamically adjust its forecasts in response to sentiment shifts observed in financial news, particularly during periods of increased market sensitivity to external events. By combining structured price data with unstructured textual sentiment, the Dana architecture improves forecast accuracy and robustness in the presence of news-driven volatility. In addition to generating predicted stock prices, the system also computes supplementary metrics such as confidence intervals, risk assessments, and volatility indicators to support more informed investment decisions.

Moreover, the application offers users an additional service of building the stock portfolio. Users will input investment amount, initial capital and desired risk score. Then the model will construct a stock portfolio tailored to user-defined investment parameters using historical return data and Particle Swarm Optimization (PSO). The system is designed to help individual investors achieve long-term financial goals by recommending an optimal asset allocation strategy based on return maximization.

4.2 Model Description and Mathematical Foundations

To analyze and predict stock movement, the sentimental analysis of the news about the companies will be done by employing a two-stage machine learning framework combining natural language processing (NLP) with regression modeling. The first component of the system is a sentiment classification model based on BERT (Bidirectional Encoder Representations from Transformers).

This model is fine-tuned to classify financial news into three sentiment categories: positive, neutral, and negative. Let T_i represent a financial news headline. The BERT architecture converts T_i into a sequence of contextual embeddings, which are processed through multiple transformer layers to produce a probability distribution over sentiment classes. Mathematically, this can be expressed as:

$$\hat{y}_i = \text{softmax}(f_{BERT}(T_i)) \in \mathbb{R}^3 \quad (1)$$

where f_{BERT} is the output layer of the BERT model, and $\hat{y}_i \in \mathbb{R}^3$ denotes the predicted probabilities for each class. The predicted label corresponds to the class with the highest probability, $\text{argmax}_j \hat{y}_{i,j}$.

The model is trained to minimize the cross-entropy loss between the predicted and actual labels,

$$L_{BERT} = -\sum_{i=1}^N \sum_{j=1}^3 y_{i,j} \log(\hat{y}_{i,j}), \quad (2)$$

where $\hat{y}_{i,j}$ is the true label for instance i .

The second component of the system is a Multilayer Perceptron (MLP) Regressor, which is designed to predict stock closing prices. The input to the MLP consists of numerical features (such as open, high, low prices and volume) as well as the sentiment label predicted by BERT.

Let $x_i \in \mathbb{R}^d$ be the feature vector for instance i , and $y_i \in \mathbb{R}^d$ be the target closing price. The MLP is a feedforward neural network that applies a sequence of linear transformations and nonlinear activations.

The hidden layers are defined as

$$\begin{aligned} h^{(1)} &= \text{ReLU}(W^{(1)}x_i + b^{(1)}), \\ h^{(2)} &= \text{ReLU}(W^{(2)}h^{(1)} + b^{(2)}), \\ \hat{y}_i &= W^{(3)}h^{(2)} + b^{(3)}, \end{aligned} \quad (3)$$

where ReLU is the ReLU activation function, $W^{(l)}, h^{(l)}$ are learnable weights and biases for layer l , $\hat{y}_i$ is the predicted stock closing price.

The training objective is to minimize the Mean Squared Error (MSE) between the predicted and actual closing prices:

$$L_{MLP} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (4)$$

Model performance is further evaluated using metrics such as Root Mean Squared Error (RMSE), defined as $\sqrt{MSE}$, and Mean Absolute Error (MAE), given by:

$$\frac{1}{N} \sum |y_i - \hat{y}_i| \quad (5)$$

This hybrid architecture leverages both textual and numerical signals to deliver more context-aware and accurate financial predictions.

Additionally, stock portfolio building is complete pipeline using historical return data and optimization through Particle Swarm Optimization (PSO). It is designed for individual investors seeking to grow their wealth by making informed, data-driven allocation decisions based on past performance.

To estimate the annualized return for each asset, the code employs the compound annual growth rate formula. For each stock i , the annual return is computed as:

$$R_i = \left(\frac{P_{end,i}}{P_{start,i}} \right)^{\frac{1}{T_i}} - 1, \quad (6)$$

where $P_{end,i}$ and $P_{start,i}$ are the stock's prices at the start and end of the dataset, and T_i represents the time period in years between the two prices, calculated from the number of calendar days. This process yields a dictionary of expected annual returns that serves as the core input for the subsequent optimization step.

The user is assumed to provide an initial capital investment and a monthly contribution over time period in months. The objective is to determine the optimal portfolio weights $w = [w_1, w_2, \dots, w_n]$, that maximize the total capital at the end of the investment period.

The expected annual portfolio return is computed as:

$$R_{portfolio} = \sum_{i=1}^n w_i * R_i, \quad (7)$$

Assuming continuous compounding with reinvested monthly contributions, the future value of the investor's portfolio after t months is approximate as:

$$V(t) = C_0 * (1 + R_{portfolio})^{t/12} + I * t * (1 + R_{portfolio})^{t/12}, \quad (8)$$

where C_0 is the initial capital and $I * t$ is the total of monthly contributions by month t . The objective function used in the PSO algorithm is therefore:

$$f(w) = -V(t) = -\left[C_0 * (1 + R_{portfolio})^{T/12} + I * t * (1 + R_{portfolio})^{T/12} \right], \quad (9)$$

where T = selected time period. Since the PSO algorithm minimizes the objective function, the negative sign ensures that the maximum future value is achieved by minimizing its negative.

The optimization algorithm begins with a population of $N = 30$ particles, each representing a potential allocation vector $w_j \in \mathbb{R}^n$. Each particle has a velocity vector $v_j \in \mathbb{R}^n$ and is updated iteratively using the standard PSO update rules:

$$\begin{aligned} v_j &\leftarrow w * v_j + c_1 * r_1 * (p_j - w_j) + c_2 * r_2 * (g - w_j), \\ w_j &\leftarrow w_j + v_j, \end{aligned} \quad (10)$$

where:

- p_j is the personal best position for particle j ,
- g is the global best across all particles,
- w , c_1 and c_2 are inertia and acceleration constants,
- r_1 and r_2 are randomly sampled.

After each particle's position is updated during optimization, the values are adjusted to stay within valid limits and ensure that the portfolio weights add up to 100%. This process is repeated for 100 iterations. At the end, the best-performing set of weights found during the process is chosen as the final recommended investment allocation.

Using these final portfolio weights, the system then simulates how the investor's capital would grow over a 36-month period. It calculates the total wealth for each month, taking into account both the initial capital and the monthly contributions. These results are stored in a time series and visualized in a graph to provide a clear view of how the investment is expected to grow over time.

The output consists of a table showing the estimated annual return for each stock and how much of the portfolio should be invested in each one. The capital growth graph is also labeled with milestone values every six months, helping the investor see the progression of their wealth.

4.3 ML Models Results

The dataset spans from 27th of January 2020 to 31th of March 2025 and includes Daily historical stock prices retrieved from Yahoo Finance, News sentiment labels extracted from manually labeled financial headlines, also ESG rating of the companies. The whole data is split using an 80/20 train-test ratio and a sliding window technique for sequence input. Both models were trained on the test sets firstly.

BERT Model were used to classify the news sentiment among test set. MLP Model was used to predict the stock prices of the companies for week from 1th to 7th of April 2025. The evaluation metrics are shown in Table 1.

Company/ Index	BERT Model				MLP Model			
	Accuracy	F1	Precision	Recall	MSE	RMSE	MAE	Accuracy
KASE	99,07%	0,9906	0,9908	0,9907	8079,75	89,88	85,17	98,48%
HSBK	96,85%	0,9628	0,9572	0,9685	1014,04	31,84	26,57	91,09%
KSPI	98,89%	0,9890	0,9892	0,9890	8697423,27	2949,13	2753,29	94,05%
KEGC	98,82%	0,9823	0,9765	0,9882	22,56	4,75	3,87	99,74%
KZAP	99,60%	0,9960	0,9961	0,9960	2001163,93	1414,62	1386,66	92,03%

Table 1. ML Models results on News Sentimental Analysis and Stock Predicition

4.4 Application Integration

The developed mobile application provides a user-centric interface that simplifies the process of stock market monitoring and personal investment management. As shown in the screenshots, the application is designed to support both novice and experienced investors through intuitive navigation and visually informative components.

The main page (Screen 1 of Figure 7) displays major stock market indices including KASE, NASDAQ, and S&P 500, each accompanied by performance metrics and a dynamic line chart reflecting historical trends over selectable timeframes (5d to 5y). Below this section, a prominently placed "Build portfolio" button encourages users to engage in personalized investment strategies. The "Favorites" area highlights selected stocks such as Apple Inc., AO Kaspi.kz, and KEGOC, showing their live prices, trading volumes, and daily change percentages, allowing for quick access and tracking.

In the market sentiment tab (Screen 2 of Figure 7), the application incorporates a "Greed and Fear Index" to visualize the prevailing investor mood. This gauge-based interface categorizes the score into fear, neutral, or greed zones, with the example score of 28 falling into the "Fear" zone—an indicator typically associated with bearish market behavior. The sentiment analysis is supported by machine learning models that aggregate and interpret financial news, helping users better understand market psychology.

The stocks page (Screen 3 and 4 of Figure 7) provides a categorized and scrollable view of available equities. Stocks are divided into sections: Local (e.g., AO Kaspi.kz, KMGZ, Halyk Bank), International (e.g., Amazon, Qualcomm, Tesla), and Popular. Each listing includes the company ticker, exchange (e.g., KASE or NASDAQ), current price, market capitalization, and performance change, enabling users to explore diverse investment options.

Additional features integrated into the app include:

- A search and filter bar for efficient asset discovery by name, sector, or region.
- Real-time news sentiment integration, where headlines are automatically tagged based on their positivity or negativity.
- Visual risk indicators and stock forecasts generated via machine learning techniques.
- Secure user interaction through encrypted communication channels and robust authentication mechanisms.

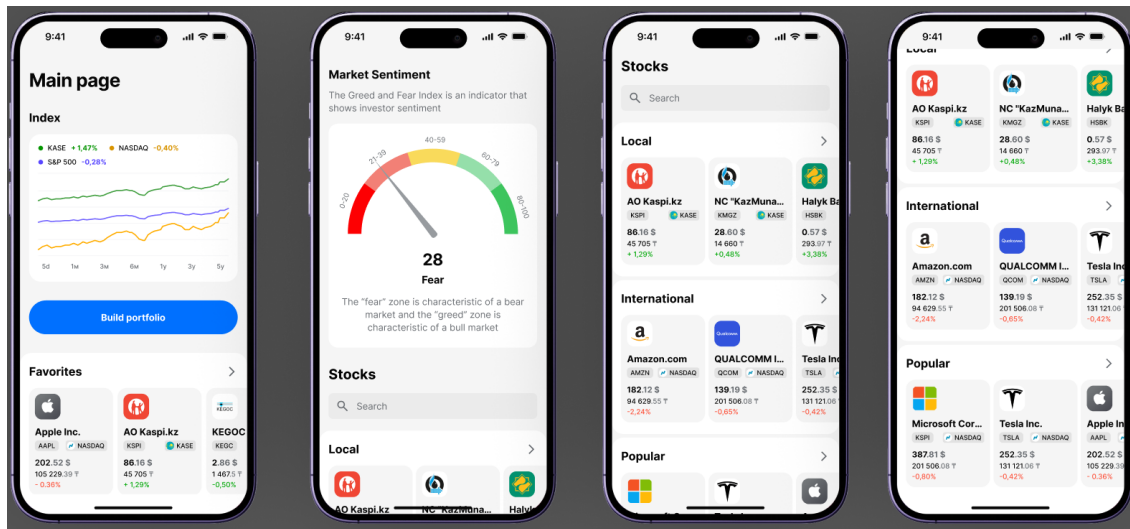


Figure 7. Mobile Application Visual Design

Future developments will focus on embedding personalized recommendations based on user financial goals, enabling direct trading through broker APIs, and providing educational content tailored to beginners. This intelligent mobile platform not only broadens access to investing tools in the local and international markets but also empowers users with insights drawn from both financial data and artificial intelligence.

5. Discussion

The number of investment applications available in app stores has been steadily increasing, particularly from fintech startups and traditional financial institutions. These platforms are often promoted as simpler and more affordable alternatives to classic brokerages, offering intuitive design and lower fees. However, as shown in Figure 8, while these apps cover various functions—from budgeting to investment access, their wealth management recommendations tend to lack depth, personalization, and advanced comparison tools.

A promising area for improvement is the use of sentiment analysis—identifying the emotional tone in financial news and social media. This technique can offer investors a clearer picture of market behavior, but it remains largely untapped in local apps. Incorporating sentiment insights, along with AI-based personalization and goal planning tools, could significantly improve user support.

This study introduced the core architecture of ECO InvestMind AI, and several directions are planned for future development. First, we aim to improve the user interface, making it more intuitive and visually engaging—especially for beginners. Clearer visuals and a simplified layout can help users better understand their financial position and feel more confident in decision-making.

We also plan to expand the news section, allowing users to browse financial headlines that are automatically tagged with sentiment labels. This will help them quickly assess market sentiment and respond accordingly. Another important focus is creating smarter, more personalized recommendations. By analyzing user behavior and financial patterns, the platform could offer dynamic strategies that evolve with the user's financial goals. Additionally, we aim to support goal-based planning, helping users link their personal milestones—like saving for education or retirement—to actionable investment steps.

App	Type	Wealth Management Recommendation Offered	Notes / Limitations
CoinKeeper	Expense & Budgeting	- Visual budgeting tools - Goal setting (e.g., savings goals)	Focus is on personal finance, not investment advice
Tabys	Entry-level Investing (AIX)	- Guidance on low-risk instruments (e.g., bonds) - Financial glossary	Does not provide personalized investment advice or portfolio suggestions
Halyk Finance	Local & Global Investment	- Access to research reports - Recommended securities list (KASE, AIX, global)	No AI-based recommendations, limited personalization
Jusan invest	Local Investment	- Suggested investment ideas - Model portfolios (e.g., conservative, balanced)	Recommendations are general, not behaviorally personalized
Freedom Broker	Global Markets Access	- Research reports and market commentary - Suggested ETFs and popular stocks	Mainly informative, no tailored wealth planning tools
Paidax	Simplified Investment Access	- Education-based recommendations - User-friendly explanations of assets	Focus on accessibility; lacks advanced portfolio strategies

Figure 8. Limitations of Existing Apps

At the same time, we acknowledge that the system has not yet been tested under high-load conditions. As we move forward, it will be essential to evaluate how the platform performs in a large-scale, decentralized environment with a high number of users and requests. Identifying potential failure cases and optimizing the system for scalability will be a key part of the next development phase.

Finally, future updates will gradually broaden the platform's coverage and include features for different investor profiles, supported by educational content tailored to various experience levels. These enhancements are intended to make the app more powerful and human-centered, helping users grow their financial knowledge and act with greater confidence.

6. Conclusion

The app uses an algorithm that is based on machine learning, which can analyze sentiment and eliminate market "noise." Using a full assessment of major interest rates, market indicators, and the emotional reaction of the audience, users are given actionable advice. This means that they save time, remain relaxed, and make better-informed decisions.

A non-responsive investment portfolio is like stagnant water. The capability of ML algorithms, combined with mobile technology, enables users to "move"—reacting rapidly to emerging opportunities and dangers. Besides technical innovation, ECO InvestMind AI solves a more substantial social problem: the absence of affordable advisory services and financial literacy in Kazakhstan and other emerging economies. By blending advanced forecasting algorithms, real-time sentiment analysis, and personalized user experience, our solution allows users to control their financial future—without requiring much investment knowledge.

While most existing platforms are transactional, ECO InvestMind AI is decision support oriented. It not only suggests what stocks to explore but also reasons why through explainable AI features. This transparency increases user confidence and stimulates trust in long-term investment schemes. In the coming future, the system will continue to evolve. Features planned to develop include in-app trading brokerage API integration, user interaction enhancing behavior analytics, and ESG-conscious recommendation engines for socially responsible investment. All these developments planned for the coming future are meant to make ECO InvestMind AI not only a utility, but an interactive digital companion to intelligent and responsible wealth generation.

In summary, our research and development work demonstrate how AI can democratize access to financial services, offering both technological robustness and real-world applicability. The model's architecture and application design serve as a scalable blueprint for future fintech solutions seeking to bridge the gap between data, insight, and user empowerment.

Acknowledgements

We would like to extend our sincere thanks to Mr. Olivier Jaylet, Senior Lecturer of the School of Information Technology and Engineering at KBTU, for his support and insightful critiques about our research journey. His deep knowledge of Artificial Intelligence and Machine Learning shaped our ML system Dana.

Our appreciation also goes to Ms. Laura Zhunusbay, UX/UI designer and friend, for her high skills, professionalism and responsiveness in teamwork. Her calmness and attention to details enhanced the quality of our project.

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